

Цель работы: сформировать умение анализировать работу протоколов транспортного уровня TCP/IP (TCP, UDP) с использованием команды «netstat».

Выполнение заданий: Вариант 11

Netstat

WindowsServer2019 [Работает] - Oracle VM VirtualBox

Файл Машина Вид Ввод Устройства Справка

Администратор: C:\Windows\system32\cmd.exe

Microsoft Windows [Version 10.0.17763.1490]
(c) Корпорация Майкрософт (Microsoft Corporation), 2018. Все права защищены.

C:\Users\Администратор>netstat -a

Активные подключения

Имя	Локальный адрес	Внешний адрес	Состояние
TCP	0.0.0.0:80	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:135	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:445	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:5985	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:47001	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:49664	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:49665	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:49666	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:49667	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:49668	MikroGoodvin:0	LISTENING
TCP	0.0.0.0:49669	MikroGoodvin:0	LISTENING
TCP	192.168.0.10:139	MikroGoodvin:0	LISTENING
TCP	:::80	MikroGoodvin:0	LISTENING
TCP	:::135	MikroGoodvin:0	LISTENING
TCP	:::445	MikroGoodvin:0	LISTENING
TCP	:::5985	MikroGoodvin:0	LISTENING
TCP	:::47001	MikroGoodvin:0	LISTENING
TCP	:::49664	MikroGoodvin:0	LISTENING
TCP	:::49665	MikroGoodvin:0	LISTENING
TCP	:::49666	MikroGoodvin:0	LISTENING
TCP	:::49667	MikroGoodvin:0	LISTENING
TCP	:::49668	MikroGoodvin:0	LISTENING
TCP	:::49669	MikroGoodvin:0	LISTENING
UDP	0.0.0.0:123	:*:	
UDP	0.0.0.0:5353	:*:	
UDP	0.0.0.0:5355	:*:	
UDP	127.0.0.1:60984	:*:	

12:02 16.10.2022

WindowsServer2019 [Работает] - Oracle VM VirtualBox

Файл Машина Вид Ввод Устройства Справка

Администратор: C:\Windows\system32\cmd.exe

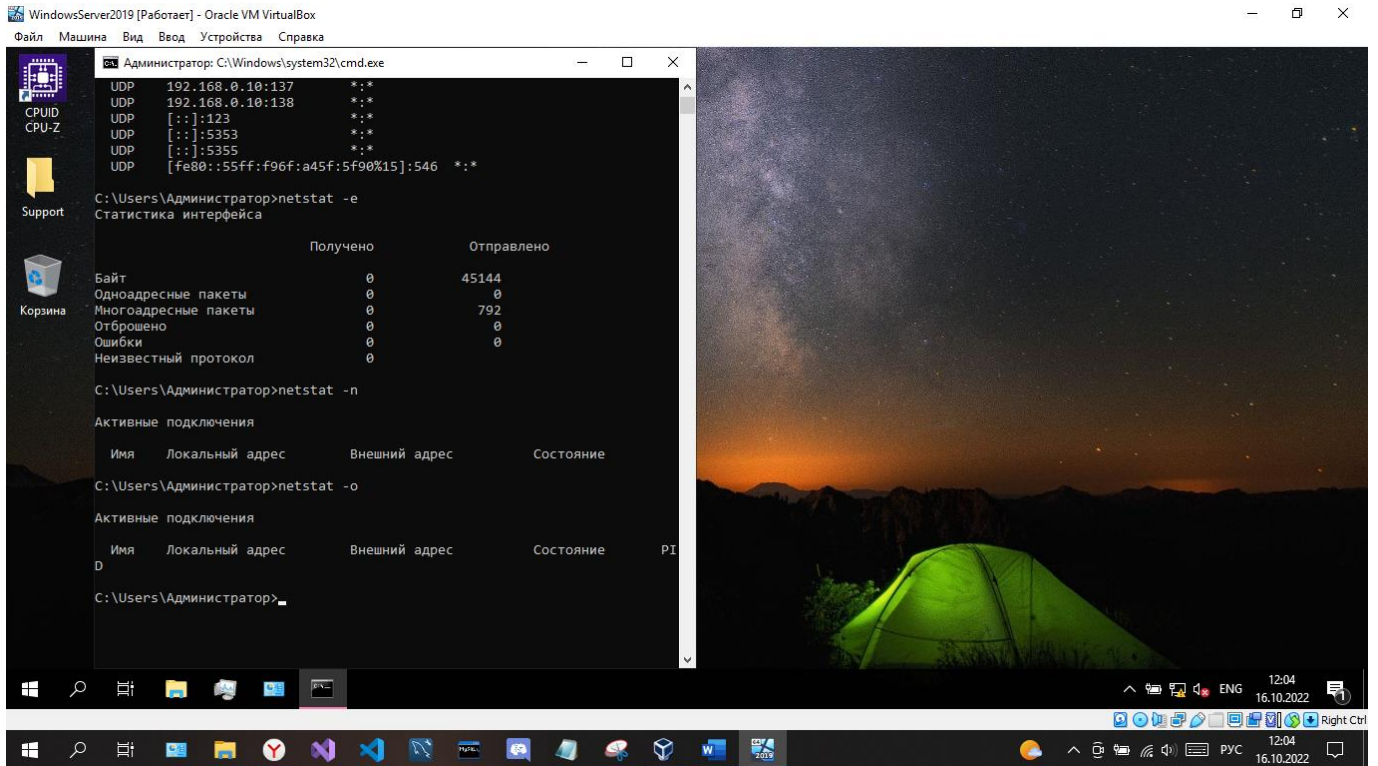
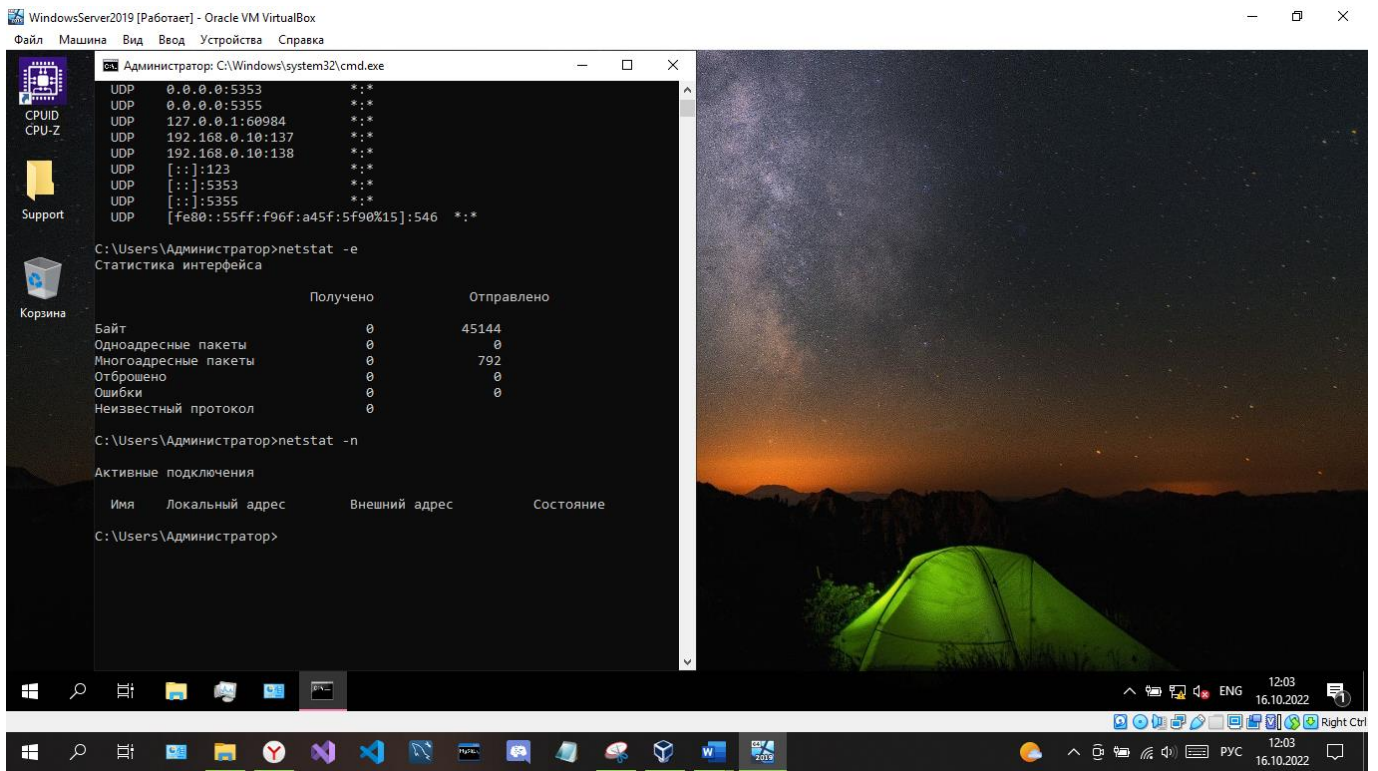
C:\Users\Администратор>netstat -e

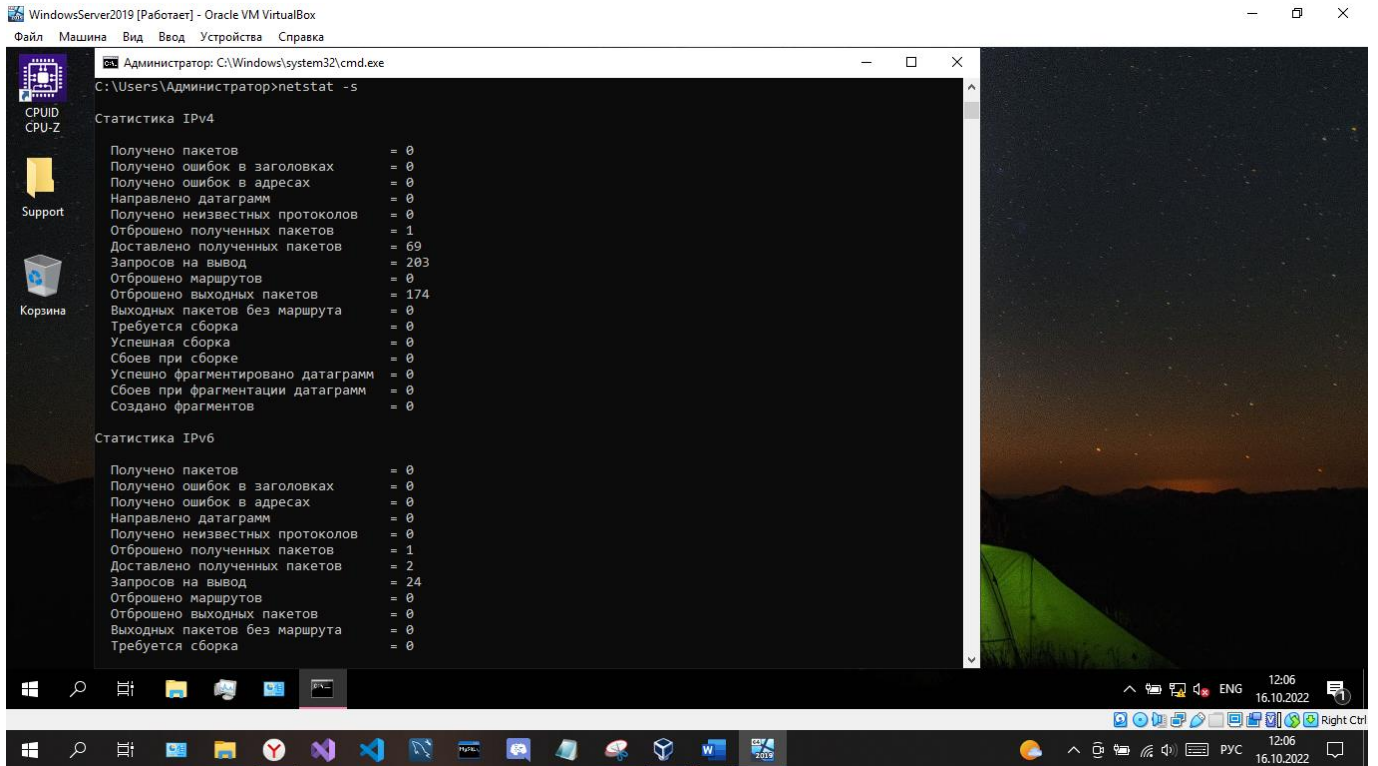
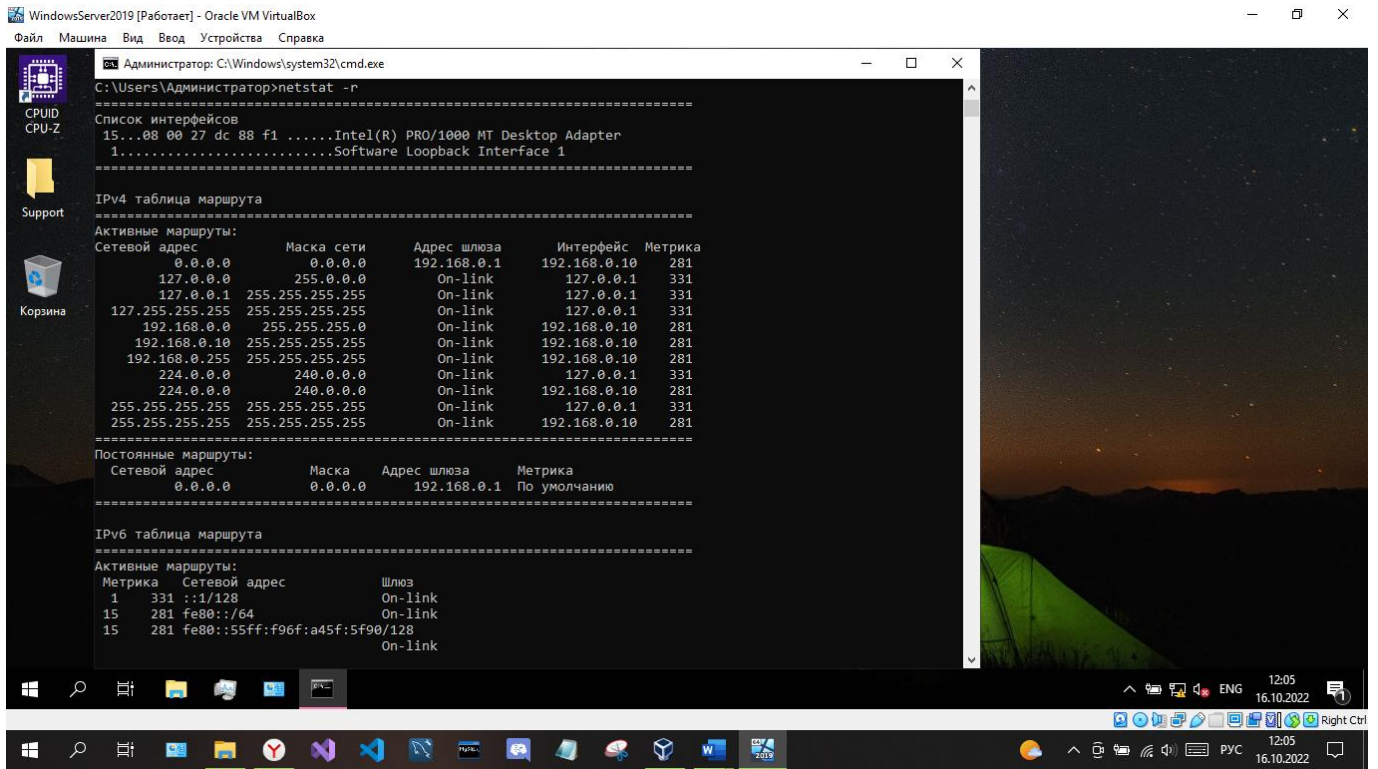
Статистика интерфейса

	Получено	Отправлено
Байт	0	45144
Одноадресные пакеты	0	0
Многоадресные пакеты	0	792
Отброшено	0	0
Ошибки	0	0
Неизвестный протокол	0	0

C:\Users\Администратор>

12:03 16.10.2022





WindowsServer2019 [Работаer] - Oracle VM VirtualBox

Файл Машина Вид Ввод Устройства Справка

Администратор: C:\Windows\system32\cmd.exe

Статистика ICMPv4

	Получено	Отправлено
Сообщений	46	46
Ошибок	0	0
'Назначение недоступно'	46	46
Превышений времени	0	0
Ошибок в параметрах	0	0
Просьба "снизить скорость"	0	0
Переадресовано	0	0
Ответных пакетов	0	0
Эхо-сообщений	0	0
Отметок времени	0	0
Ответы на отметки времени	0	0
Масок адресов	0	0
Ответов на маски адресов	0	0
Маршрутизатор	0	0
Маршрутизатор	0	0

ICMPv6 Статистика

	Получено	Отправлено
Сообщений	0	5
Ошибок	0	0
'Назначение недоступно'	0	0
Пакет слишком велик	0	0
Превышений времени	0	0
Ошибок в параметрах	0	0
Эхо-сообщений	0	0
Ответных пакетов	0	0
MLD-запросы	0	0
MLD-отчеты	0	0
MLD выполнено	0	0
Маршрутизатор	0	3
Маршрутизатор	0	0
Окружение	0	1

12:06 16.10.2022

Access-list

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical 814 y: 202

Root 02:34:30

Time: 00:05:06

Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit

Copper Cross-Over

20:11 12.09.2022

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name VLAN4
Switch(config-vlan)#exit
Switch(config)#vlan 5
Switch(config-vlan)#name VLAN5
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

☐ Top



Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config-vlan)#name VLAN4
Switch(config-vlan)#exit
Switch(config)#vlan 5
Switch(config-vlan)#name VLAN5
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

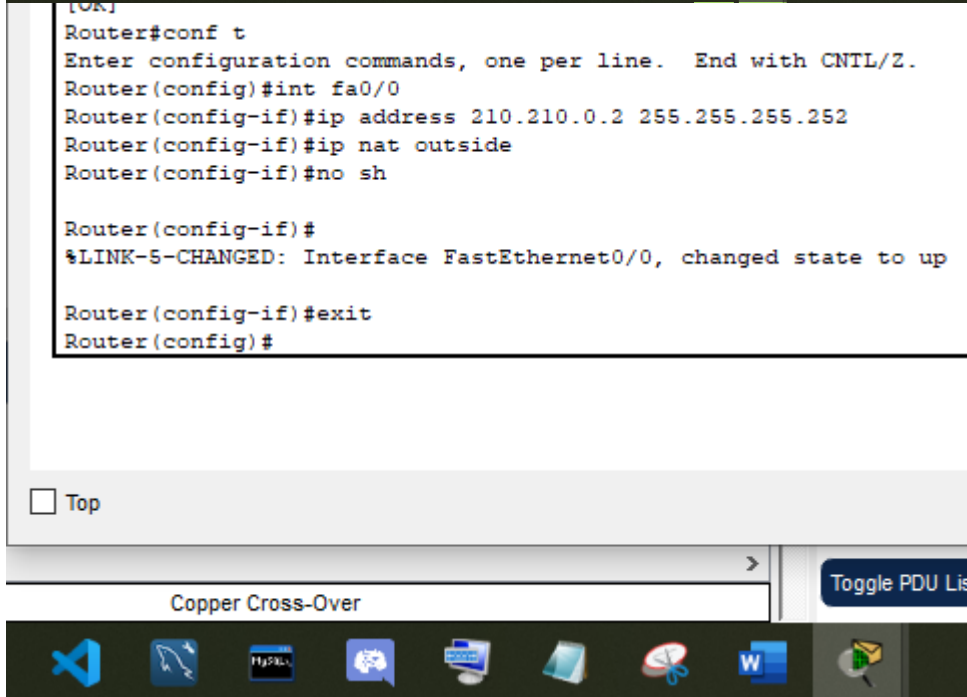
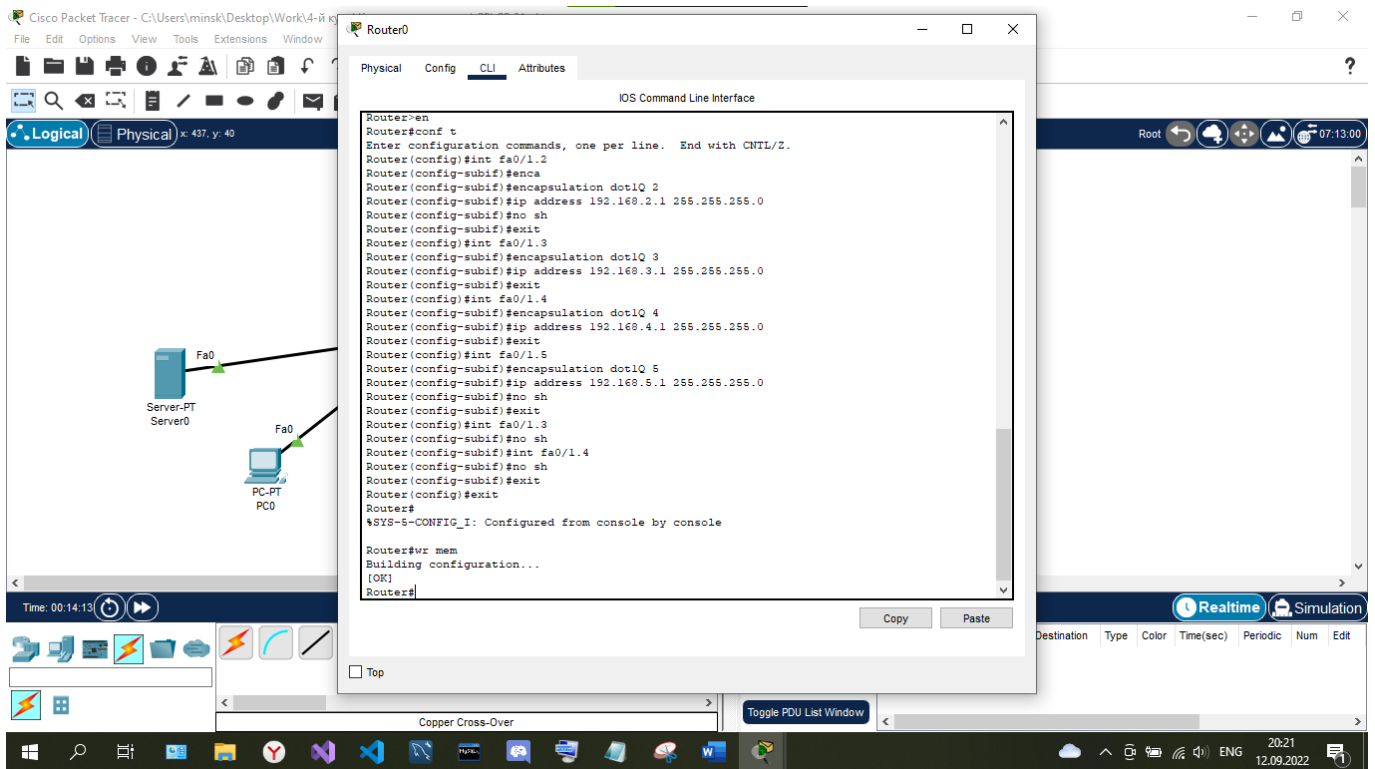
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#int fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 5
Switch(config-if)#exit
Switch(config)#int fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3,4,5
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

☐ Top

Copy Paste

20:16 12.09.2022 ENG



```
[OK]
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/24
Switch(config-if)#no sh
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

☐ Top

Copper Cross-Over

Toggle PD

```
Router(config)#int fa0/1
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1.3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1.4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1.5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Copy

Past

☐ Top

Copper Cross-Over

Toggle PDU List Window

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 716, y: 128

Server-PT Server0

PC-PT PC0

Time: 00:22:39

Router1

Physical Config CLI Attributes

IOS Command Line Interface

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#int fa0/0

Router(config-if)#no sh

Router(config-if)#

%LINK-S-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#int fa0/1

Router(config-if)#no sh

Router(config-if)#

%LINK-S-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#

Copy Paste

Top

Copper Cross-Over

Toggle PDU List Window

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 716, y: 128

Server-PT Server0

PC-PT PC0

Time: 00:24:29

Router1

Physical Config CLI Attributes

IOS Command Line Interface

Router(config)#int fa0/1

Router(config-if)#no sh

Router(config-if)#

%LINK-S-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#int fa0/0

Router(config-if)#ip address 210.210.0.1 255.255.255.252

Router(config-if)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#int fa0/0

Router(config-if)#ip address 210.210.0.1 255.255.255.252

Router(config-if)#exit

Router(config)#int fa0/1

Router(config-if)#ip address 210.210.1.1 255.255.255.252

Router(config-if)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem

Building configuration...

[OK]

Router#

Copy Paste

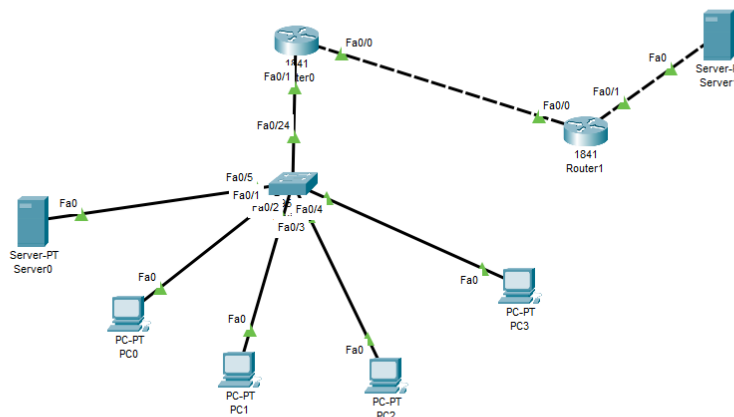
Top

Copper Cross-Over

Toggle PDU List Window

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit



Time: 00:24:42

Realtime Simulation



Copper Cross-Over

Scenario 0

New Delete

Toggle PDU List Window

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int range fa0/1.2
Router(config-if-range)#ip nat inside
Router(config-if-range)#exit
Router(config)#int range fa0/1.3
Router(config-if-range)#ip nat inside
Router(config-if-range)#exit
Router(config)#int range fa0/1.4
Router(config-if-range)#ip nat inside
Router(config-if-range)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

☐ Top

Copper Cross-Over

Toggle PDU List Window



```

Router(config)#ip access-list standard FOR-NAT
Router(config-std-nacl)#permit 192.168.2.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.3.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.4.0 0.0.0.255
Router(config-std-nacl)#exit
Router(config)#ip nat inside source list FOR-NAT interface fa0/0 overload
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 210.210.0.1
Router(config)#ip nat inside source list FOR-NAT interface fastEthernet 0/0 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#

```

Copy

☐ Top

Copper Cross-Over

Toggle PDU List Window

The screenshot shows the Cisco Packet Tracer interface. In the background, a network diagram is visible with a 'Server-PT Server0' and a 'PC-PT PC0' connected via 'Fa0' interfaces. A 'Copper Cross-Over' cable is shown between them. The 'Logical' tab is selected, and the network is in 'Realtime' mode. The time is 00:43:44.

In the foreground, a 'Command Prompt' window is open, displaying the following output:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 210.210.1.2

Pinging 210.210.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 210.210.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

The command prompt window also shows a 'Toggle PDU List Window' button at the bottom.

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 319, y: 198

Server-PT Server0

PC-PT PC0

Time: 00:44:48

Server0

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 210.210.1.2

Pinging 210.210.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 210.210.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

Copper Cross-Over

Toggle PDU List Window

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 714, y: 129

Server-PT Server0

PC-PT PC0

Time: 00:47:01

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.0.0 255.255.0.0 210.210.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 192.168.5.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.5.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#ping 192.168.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Copy Paste

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

Copper Cross-Over

Toggle PDU List Window

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 418, y: 45

Time: 00:49:38

Server-PT Server0

PC-PT PC0

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip access-list extended FROM-OUTSIDE
Router(config-ext-nacl)#deny ?
  ahp      Authentication Header Protocol
  eigrp    Cisco's EIGRP routing protocol
  esp      Encapsulation Security Payload
  gre      Cisco's GRE tunneling
  icmp     Internet Control Message Protocol
  ip       Any Internet Protocol
  ospf     OSPF routing protocol
  tcp      Transmission Control Protocol
  udp      User Datagram Protocol
Router(config-ext-nacl)#deny ip any 192.168.2.0 0.0.0.255
Router(config-ext-nacl)#deny ip any 192.168.3.0 0.0.0.255
Router(config-ext-nacl)#deny ip any 192.168.4.0 0.0.0.255
Router(config-ext-nacl)#deny ip any 192.168.5.0 0.0.0.255
Router(config-ext-nacl)#exit
Router(config)#int fa0/0
Router(config-if)#ip access-group FROM-OUTSIDE in
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh mem
Building configuration...
[OK]
Router#
```

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Toggle PDU List Window

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

Copper Cross-Over

20:57 12.09.2022

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 707, y: 135

Time: 00:51:15

Server-PT Server0

PC-PT PC0

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#ping 192.168.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.5.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.5.2, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)

Router#ping 192.168.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)

Router#ping 192.168.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)

Router#
```

Copy Paste

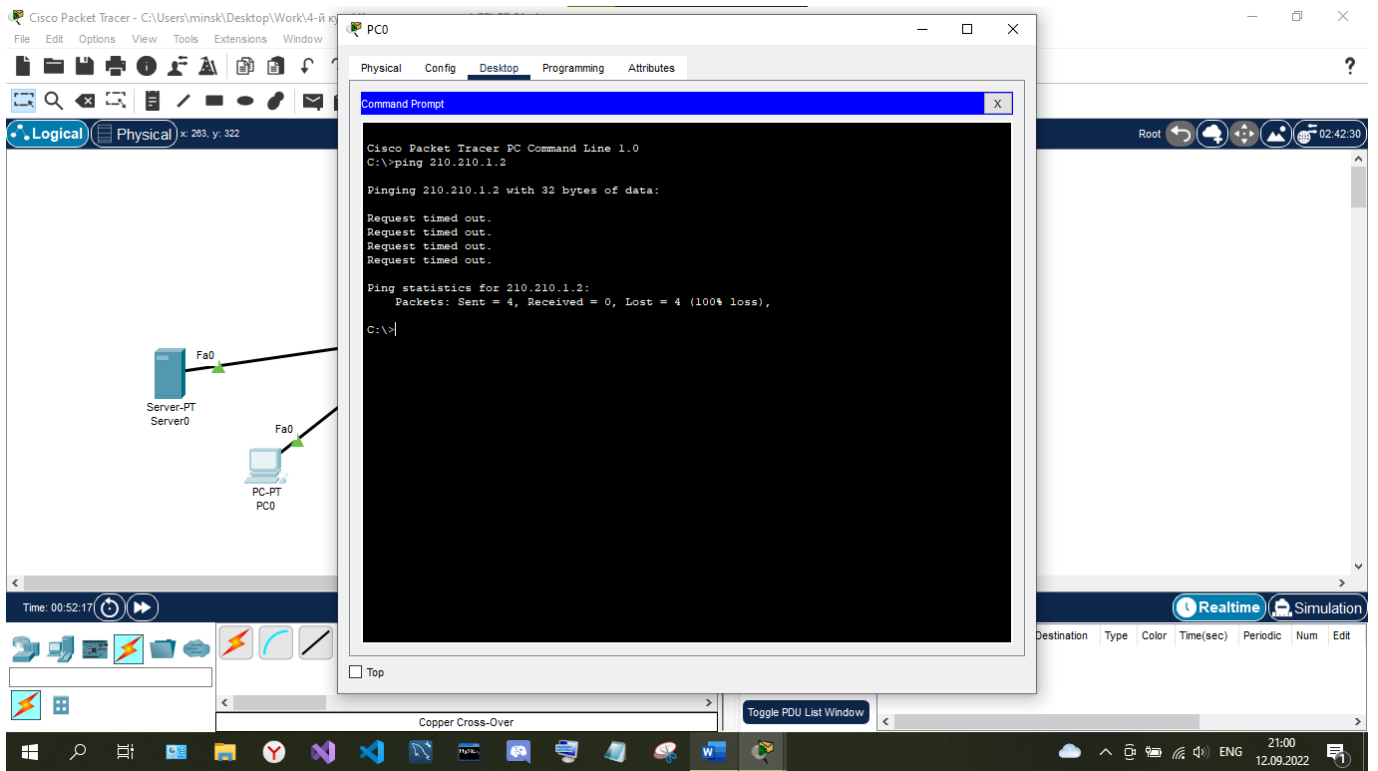
Toggle PDU List Window

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

Copper Cross-Over

20:59 12.09.2022



```
!
ip access-list standard FOR-NAT
  permit 192.168.2.0 0.0.0.255
  permit 192.168.3.0 0.0.0.255
  permit 192.168.4.0 0.0.0.255
ip access-list extended FROM-OUTSIDE
  deny ip any 192.168.2.0 0.0.0.255
  deny ip any 192.168.3.0 0.0.0.255
  deny ip any 192.168.4.0 0.0.0.255
  deny ip any 192.168.5.0 0.0.0.255
!
!
!
!
--More--
```

☐ Top

Copper Cross-Over



```
% Invalid input detected at '^' marker.

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip access-list extended FROM-OUTSIDE
Router(config-ext-nacl)#permit ip any host 210.210.0.2
Router(config-ext-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

☐ Top

Copper Cross-Over

Toggle PDU Lis

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 316, y: 302

Time: 01:21:36

PC Laptop Server

PC0

Server-PT
Server0

PC-PT
PC0

Fa0

Fa0

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 210.210.1.2

Pinging 210.210.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 210.210.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 210.210.1.2

Pinging 210.210.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 210.210.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

☐ Top

Root

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

Copper Cross-Over

Toggle PDU List Window

Windows taskbar: 21:30 12.09.2022

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip access-list extended FROM-OUTSIDE
Router(config)#ip access-list extended FROM-OUTSIDE
Router(config-ext-nacl)#permit ip any host 210.210.0.2
Router(config-ext-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

☐ Top

Copper Cross-Over

Toggle PDU List Window

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-4-18-2022\4-4-18-2022

File Edit Options View Tools Extensions Window

Logical Physical x: 578, y: 421

Time: 01:23:50

Server-PT Server0

PC-PT PC0

PC0

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 210.210.1.2

Pinging 210.210.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 210.210.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 210.210.1.2

Pinging 210.210.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 210.210.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
```

Root

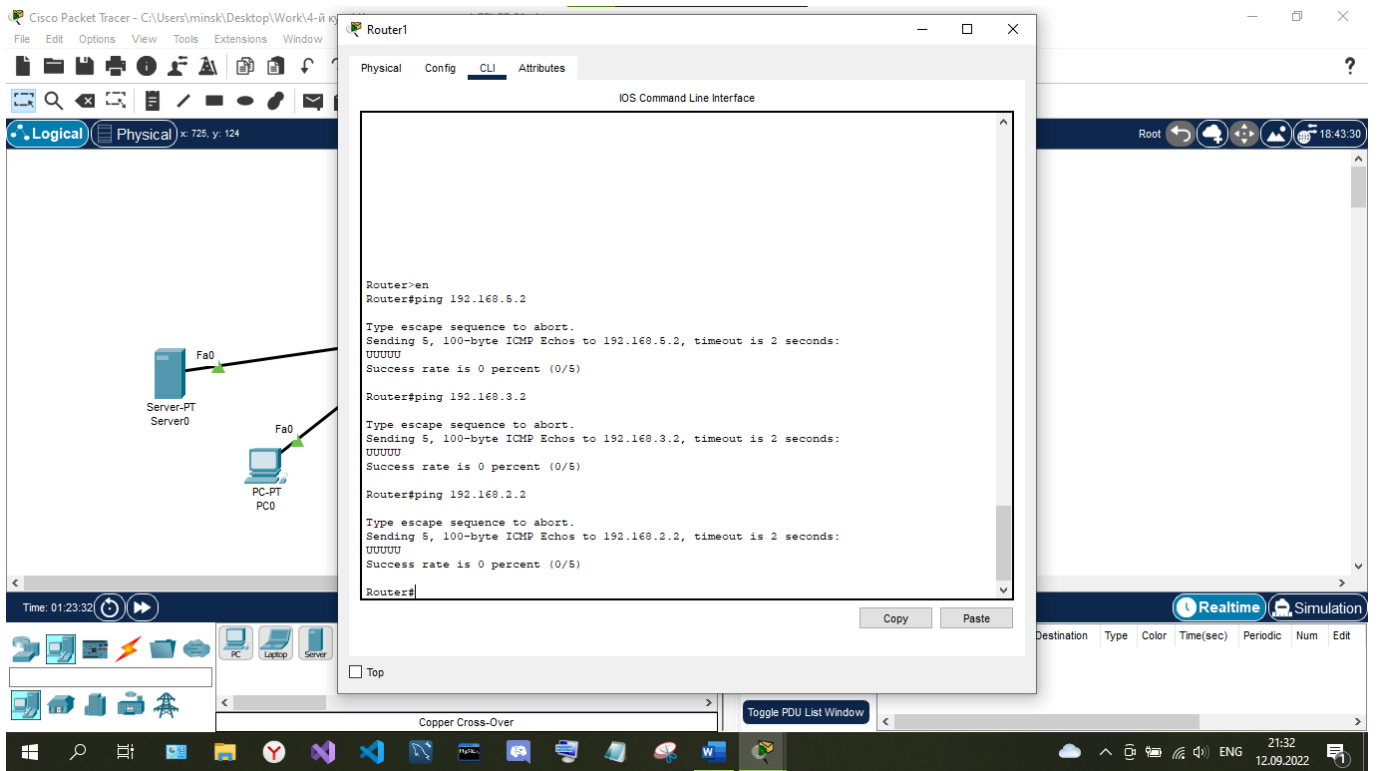
Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

Copper Cross-Over

Toggle PDU List Window

21:32 12.09.2022



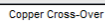
```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#username admin privilege 15 password cisco
Router(config)#enable password cisco
Router(config)#line vty 0 4
Router(config-line)#login LOCAL
Router(config-line)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip access-list extended FROM-OUTSIDE
Router(config-ext-nacl)#deny tcp any host 210.210.0.2 eq telnet
Router(config-ext-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```


Top

Toggle PDU I



Toggle PDU List Window

```
Router(config-ext-nacl)#permit tcp 192.168.3.0 0.0.0.255 host 210.210.1.2 eq www
Router(config-ext-nacl)#permit ip host 192.168.3.100 host 210.210.1.2
Router(config-ext-nacl)#deny ip 192.168.3.0 0.0.0.255 host 210.210.1.2
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1.3
Router(config-subif)#ip access-group FROM-USERS in
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Copy

Paste

☐ Top

Copper Cross-Over

Toggle PDU List Window

Cisco Packet Tracer - C:\Users\mimsk\Desktop\Work

File Edit Options View Tools Extensions Windows

Physical Config Desktop Programming Attributes

Web Browser

< > URL http://210.210.1.2 Go Stop

Request Timeout

Server-PT Server0

PC-PT PC0

Time: 01:43:30

Realtime Simulation

Source Destination Type Color Time(sec) Periodic Num Edit

Top

Copper Cross-Over

Toggle PDU List Window

21:52 12.09.2022

```
C:\>ping 210.210.1.2
```

```
Pinging 210.210.1.2 with 32 bytes of data:
```

```
Reply from 192.168.3.1: Destination host unreachable.
```

```
Reply from 192.168.3.1: Destination host unreachable.
```

```
Reply from 192.168.3.1: Destination host unreachable.
```

```
Reply from 192.168.3.1: Destination host unreachable.
```

```
Ping statistics for 210.210.1.2:
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>|
```

☐ Top

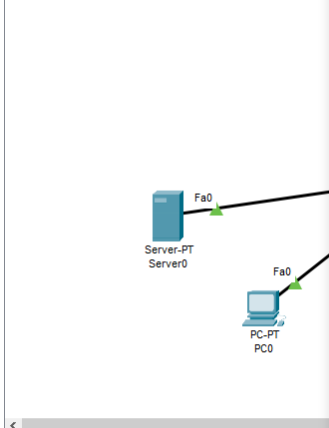
Copper Cross-Over



Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к...

File Edit Options View Tools Extensions Window

Logical Physical x: 518, y: 379



Time: 01:44:22

PC Laptop Server

Toggle PDU List Window

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 210.210.1.2

Pinging 210.210.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

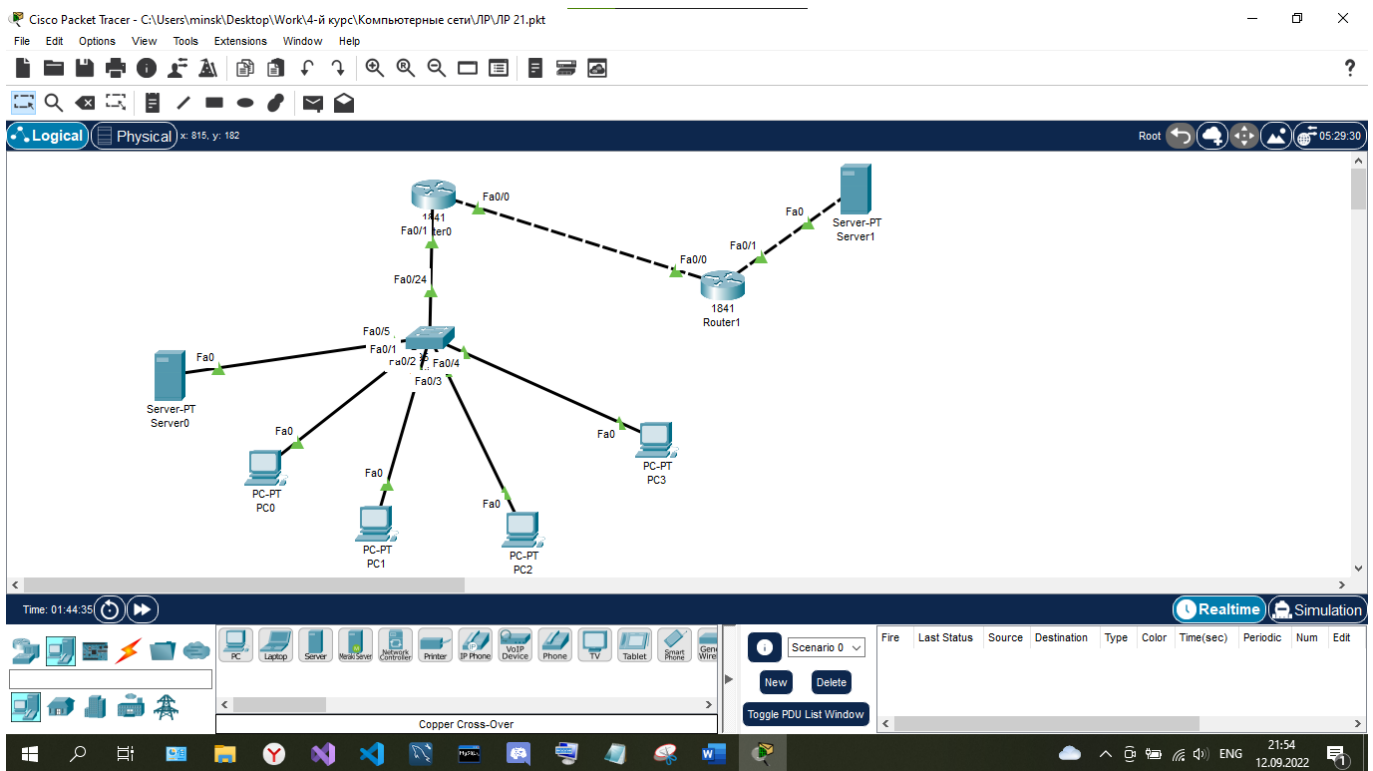
Ping statistics for 210.210.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

☐ Top

Copper Cross-Over

Toggle PDU List Window



Контрольные вопросы

- ## 1. Что такое порт?

Порт - это программное понятие, которое используется клиентом или сервером для отправки или приема сообщений; порт идентифицируется 16-битовым числом.

- ## 2. Как осуществляется адресация?

Адресация осуществляется с использованием IP адресов. Но после того как пакет средствами протокола IP доставлен в компьютер-получатель с заданным IP адресом, данные необходимо направить конкретному прикладному процессу-получателю. Каждый компьютер может выполнять несколько процессов, более того, прикладной процесс может иметь несколько точек входа, одновременно выступающих в качестве получателей данных.

- ### 3. Что значит "основанный на соединении"?

Термин "основанный на соединении" означает, что два приложения, использующие ТСР (как правило, это клиент и сервер), должны установить ТСР соединение друг с другом, после чего у них появляется возможность обмениваться данными.

- #### 4. Что такое сокет?

Сокет - это номер порта в совокупности с номером сети и номером конечного узла однозначно определяют прикладной процесс в сети.

Вывод: я научился анализировать работу протоколов транспортного уровня TCP/IP (TCP, UDP) с использованием команды «netstat».